Noun and Adjective Order in Emerging Sign Languages and Silent Gesture

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Introduction. Typological surveys reveal a cross-linguistic preference for Noun-Adjective (NounAdj) order in both spoken and signed languages [1, 2]. We investigate whether this preference reflects a cognitive bias favoring NounAdj order. Prior work on emerging sign languages and laboratory silent gesture tasks suggests cognitive defaults for argument structure (e.g., SOV) when linguistic conventions are absent [3, 4, 5]. However, evidence for a parallel bias in the ordering of nouns and adjectives in silent gesture remains inconclusive. While Culbertson et al. [6] observed NounAdj adjacency, their design targeted modifiers in complex phrases (e.g., Dem-Num-N-Adj) rather than basic ordering preferences. Similarly, Holtz et al. [7] identified a perceptual preference for NounAdj but did not test production. We explore the NounAdj ordering bias in two complementary studies: Study 1 examines natural emergence of ordering in Nicaraguan Sign Language (NSL) and Zinacantec Family Homesign (Z Sign, used by members of a single extended family in Mexico) [8]; Study 2 investigates experimental emergence in silent gesture.

Study 1: Natural Emergence. Methods (1a): We analyzed productions from 24 NSL signers (from all generational 'cohorts' of signers) using stimuli that contrast typical (e.g., hammering) vs. atypical (e.g., dropping a hammer) tool-use events [3]. We restricted analysis to *atypical* trials to strictly disambiguate object references (nouns) from representations of actions (verbs) [9]. We identified 36 strings containing one noun and one adjective-like modifier (signs that convey an attribute, e.g., size, color), produced by 14 of the 24 signers. **Results (1a):** NSL signers overwhelmingly preferred NounAdj order (86%, 31/36). An intercept-only model confirmed that NounAdj production was significantly above chance ($\beta = 1.82$, $SE = 0.48$, $p < .001$). This preference was robust across cohorts, with all signers but one producing a majority of NounAdj responses. **Methods (1b):** We applied the same methodology to 6 signers (3 deaf, 3 hearing) of Z Sign. **Results (1b):** The sample yielded 22 analyzable tokens and no clear ordering preference (NounAdj: 10/22, 45%; AdjNoun: 12/22, 55%). To compensate for the small sample size, we designed additional stimuli using the same paradigm. This larger dataset revealed a robust preference for the *opposite* order, AdjNoun (72%, 102/142; $\beta = 1.00$, $SE = 0.34$, $p = .003$).

Study 2: Experimental Emergence. Methods (2a): We analyzed silent gesture productions from 20 sign-naive, hearing English speakers [10] describing the same atypical events used in Study 1. **Results (2a):** Participants showed a robust preference for NounAdj order (88%, 29/33; $\beta = 1.98$, $p < .001$), with most participants producing NounAdj on every trial. **Methods (2b):** To validate findings using stimuli explicitly designed to elicit modifiers, we conducted a second experiment ($n = 20$) with a minimal contrast design. Participants described object pairs presented side-by-side that differed along a single visual dimension (see Figure 1). This design disambiguated object (noun) from property (adjective) gestures and tested whether NounAdj ordering is modulated by semantic type of modifier (e.g., scalar vs. absolute). **Results (2b):** The preference for NounAdj order remained robust across all semantic classes (234/269, 87%; $\beta = 3.09$, $SE = .74$, $p < .001$, see Figure 2). There were no significant differences between Size (scalar), Shape, or Texture. The only significant deviation was Pattern, which elicited an even higher rate of NounAdj productions than the reference category Size ($\beta = 2.61, p < .01$).

Conclusion and implications: NSL and silent gesture show a strong NounAdj ordering bias, consistent with a pressure to establish a referential anchor before adding attributive information. However, this tendency is not universal among emerging systems: Z Sign showed the opposite preference (AdjNoun), suggesting that language-specific conventions can override a potential cognitive NounAdj default. Our study is limited in that we could not apply syntactic tests to diagnose the syntactic function of signs: adjective-like signs may function as nouns or be embedded in relative clauses, which influences word order [2]. Overall, our findings suggest that 'Adjective' is not a primitive but an emergent category, and its order relative to the Noun crystallizes through multiple interacting forces, including cognitive pressure toward referential grounding, conventionalization of property expressions, and the status of lexical categories themselves.

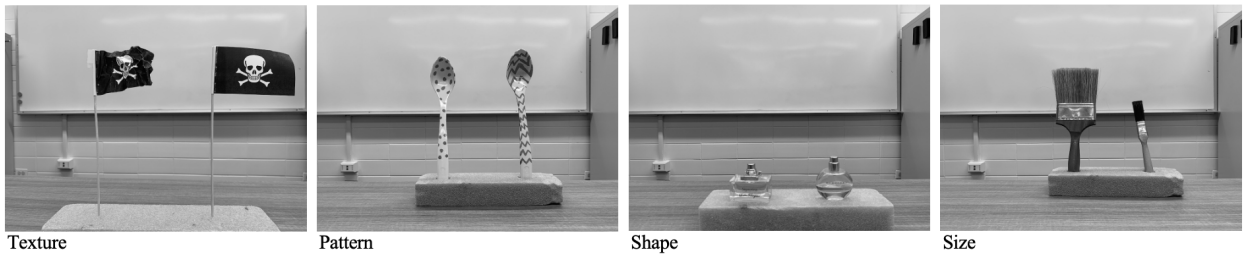

Figure 1: Sample stimuli in Study 2b

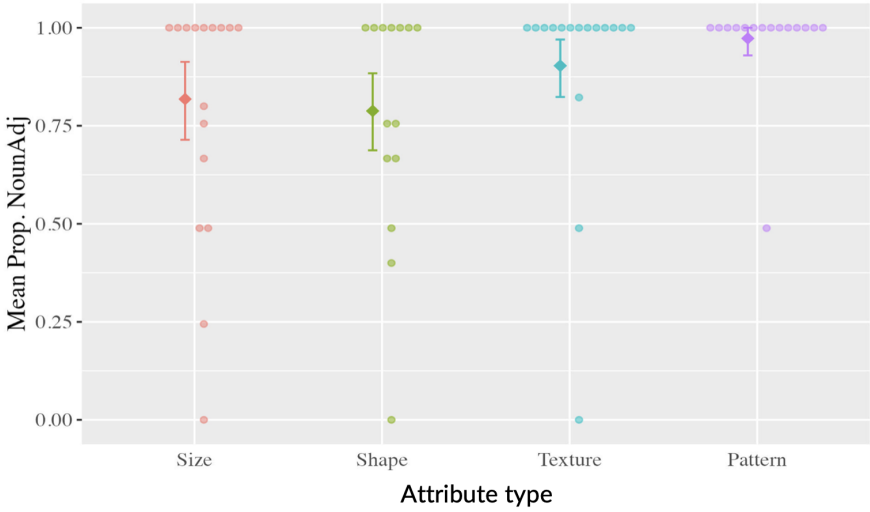

Figure 2: Mean proportion of NounAdj word orders produced in each Attribute Type condition. Dots: individual participants; Error bars: 95% CIs

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